Text & Document

COMP7507
Visualization & Visual Analytics



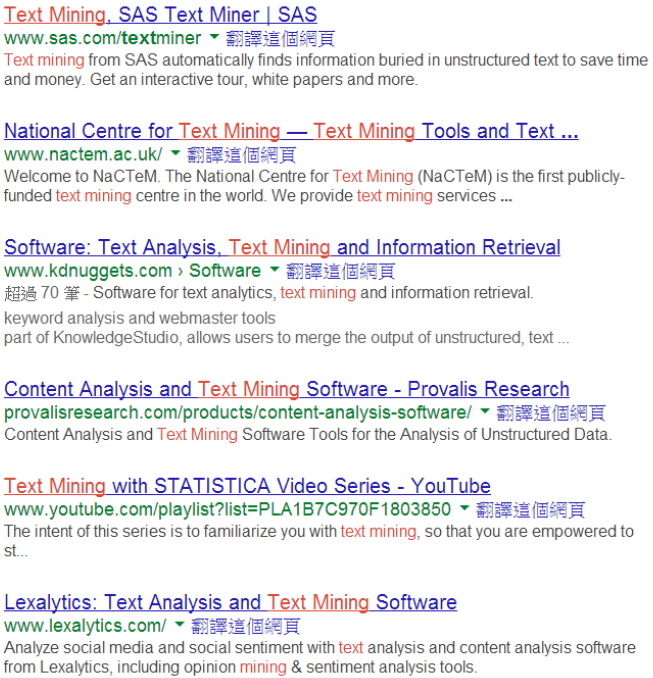
SCHOOL OF
COMPUTING &
DATA SCIENCE
The University of Hong Kong

Textual Information

- **Text** is the basic medium of communication
- Easy to get access to the ever-growing volume of documents with textual information
  - Digital libraries: E-book libraries (e.g., Google Books), Journals & magazines (e.g., IEEE, ACM, Elsevier)
  - WWW: blogs, Facebook, Twitter, webpages, Wikipedia
  - Many others (e.g., subtitles, lyrics)

Text Analytics

- How to find information from piles of text?
- Search results:



Text Analytics

Text Summarization

- To generate a precise summary of a given large amount of text information
- Example
  - From over 480K confirmed records of H1N1 (swine flu) influenza
    - Ask: What are the major sources of infection?
  - From over 5K customer reviews of your online store
    - Ask: What did the customers say about your services?

Text Analytics

Opinion Mining (aka Sentiment Analysis)

- To extract and quantify affections (feelings) and subjective information from a given large amount of textual information
- Example
  - From thousands of customer reviews
  - Ask:
    - How have your customers' opinions changed toward your services?
    - How do your customer feel about your new product?
    - What are the main features of your products that your customers like most?

Text Analytics

- Steps further:
  - How do the reviews on your products compare with your competitors'?
  - Any correlations between the blogger sentiments and movie box office?

The Challenges

- Huge volume of complex information
  - Textual data are unstructured
  - Difficult to understand the meaning
  - Hard to perform analysis
- Different customs on use of languages
- Machines alone is not good enough for full analysis

Levels of Text Analysis

1. **Lexical level:** To group a string of characters into tokens, which is the basic unit of text to be analyzed and is application dependent

- Example of tokens:
  - Words
    - A word may be defined as a sequence of letters separated by space or punctuation
    - Are apple and apples the same word?
    - Is @ a word? How about * ?) . , ?
    - Are Mister and Mr the same?
    - How many words are there in aren't, dunno ?
    - How about Chinese text? How to identify words?

兩名作家向美國三藩市聯邦法院提出訴訟，指控人工智能開發公司OpenAI，不正當使用作品，用以訓練俗稱聊天機械人的運算程式。

Levels of Text Analysis

- Lexical level (cont'd): More example of tokens
  - Word stems
    - The inflected variants of a word
    - e.g., have, has, had, having
  - Phrases
    - A group of words used as a unit to express a concept
    - e.g., cool down, keen on
  - Word n-grams
    - A sequence of *n* words
    - e.g., 2-grams: I like, I go, I eat, I think
  - Character n-grams
    - A sequence of *n* characters
    - e.g., the, con, doc, thi

Levels of Text Analysis

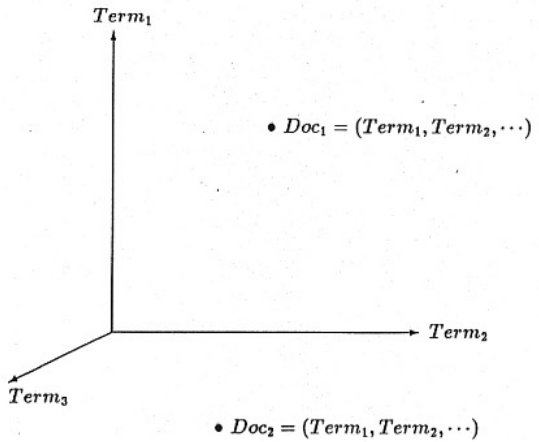
2. **Syntactic level**
   - To parse the purpose of tokens
   - Examples:
     - Grammatical category (nouns, adjectives, conjunctions, etc.)
     - Tense
     - Plurality
     - Richer purpose such as date, money, place, person, etc.
3. **Semantic level**
   - To extract meaning of the syntactic structures of the full text

Document Statistics

- Number of words, paragraphs
- Word distribution or frequencies
- With document statistics, we may ask if there is any relationship between paragraphs or documents within a corpus (i.e., a collection of documents)?

Vector Space Model

- Each term in a corpus corresponds to a dimension in an *n*-dimensional space
- Each document is a vector with the *i*-th coordinates being the weight (or importance) of the *i*-th term
- Pre-processing of documents:
  - Filtering: remove stop words (e.g., "the" or "a")
  - Stemming: group inflected forms of a word (e.g., "robot", "robots", "robotics", "robotlike")

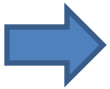


[Salton and McGill, *Introduction to Modern Information Retrieval*, 1983.]

Vector Space Model

- E.g., using frequency of word occurrences as importance

The special kind of boredom from which modern urban populations suffer is intimately bound up with their separation from the life of Earth. It makes life hot and dusty and thirsty, like a pilgrimage in the desert. Among those who are rich enough to choose their way of life, the particular brand of unendurable boredom from which they suffer is due, paradoxical as this may seem, to their fear of boredom. In flying from the fructifying kind of boredom, they fall a prey to the other far worse kind. A happy life must be to a great extent a quiet life, for it is only in an atmosphere of quiet that true joy can live.



Word	Freq.	Word	Freq.
life	5	atmosphere	1
boredom	4	live	1
kind	3	urban	1
quiet	2	extent	1
suffer	2	dusty	1
population	1	due	1
way	1	thirsty	1
unendurable	1	enough	1
pilgrimage	1	paradoxical	1
far	1	bound	1
choose	1	seem	1
special	1	worse	1
Earth	1		

http://www.wordcounter.com

Document Retrieval

- Suppose we have a query and want to retrieve the most relevant document to the query.
- Query: "who took computer science"
- Document:
  - D1: "alan turing is considered the father of computer science"
  - D2: "who published in science"
  - D3: "computer science is not just science"
  - D4: "it was alan who took me to the science class"
- We define a matching score $S_{q,d}$ for a document ***d*** to a query ***q***
- There can be different definitions for $S_{q,d}$

Term Frequency

- **Term Frequency** ($tf_{t,d}$) of term ***t*** in document ***d*** is defined as the number of times that ***t*** occurs in ***d***.
- A term has higher weight if it appears very often in a document
- Given a query (a string of words) ***q***, we may define the score of a document by

$$S_{q,d} = \sum_{t \in q} tf_{t,d}$$

The summation of the term frequencies of all words appearing in the query

Here, we consider a document as a bag of words without ordering (using the vector space model)

Term Frequency

Query: who took computer science

$$S_{q,d} = \sum_{t \in q} tf_{t,d}$$

	Documents	$tf_{t,d}$				$S_{q,d}$
		who	took	computer	science	
D1	alan turing is considered the father of computer science			1	1	2
D2	who published in science	1			1	2
D3	computer science is not just science			1	2	3
D4	it was alan who took me to the science class	1	1		1	3

Term Frequency

- However, relevance is not directly proportional to term frequency. Hence, use log scale instead:

$$S_{q,d} = \Sigma_{t \in q} \log (1 + tf_{t,d})$$

	Documents	$\log (1+tf_{t,d})$				$S_{q,d}$
		who	took	computer	science	
D1	alan turing is considered the father of computer science			0.301	0.301	0.602
D2	who published in science	0.301			0.301	0.602
D3	computer science is not just science			0.301	0.477	0.778
D4	it was alan who took me to the science class	0.301	0.301		0.301	0.903

Term Frequency

- What's the problem with using only term frequency to rank documents?
  - All terms have the same importance
- Recall the stop words? Frequently appear in documents but less informative
- Doctrine: *A term that appears in every document is less important, because it has no discriminatory power*

Inverse Document Frequency

- We would like rare terms to have higher weighting
- df_t — **document frequency** of term ***t*** is defined as the number of documents that contain ***t***
  - $df_t \leq N$, the total number of documents
  - $df_t \geq 1$, assuming that we consider only terms appearing in the corpus
- **Inverse document frequency**

$$idf_t = \log (N / df_t)$$

Inverse Document Frequency

Query: who took computer science

	Documents	Terms			
		who	took	computer	science
D1	alan turing is considered the father of computer science			1	1
D2	who published in science	1			1
D3	computer science is not just science			1	2
D4	it was alan who took me to the science class	1	1		1
	df	2	1	2	4
	idf	0.301	0.602	0.301	0

tf-idf weighting

- **Term frequency inverse document frequency**
- The tf-idf weight of a term ***t*** is the product of its tf and idf weights

$$\text{tf-idf}_{t,d} = \log(1 + \text{tf}_{t,d}) * \log(N / \text{df}_t)$$

- A term has
  - Higher weight if it has high frequency in a document
  - Lower weight if it appears in more documents

tf-idf weighting

- Matching score of document ***d*** to a query ***q*** using tf-idf weighting :

$$S_{q,d} = \Sigma_{t \in q} \text{tf-idf}_{t,d}$$

- Increases with the number of matching query term occurrences **within** a document
- Increases with the rarity of a matching query term **across** the whole corpus

Document Retrieval with tf-idf

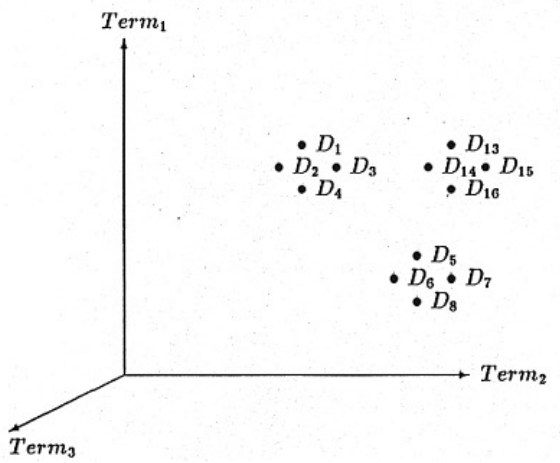
Query: who took computer science

	Documents	$\text{tf-idf}_{t,d}$				$S_{q,d}$
		who	took	computer	science	
D1	alan turing is considered the father of computer science			0.09	0	0.09
D2	who published in science	0.09			0	0.09
D3	computer science is not just science			0.09	0	0.09
D4	it was alan who took me to the science class	0.09	0.18		0	0.27

Document Clustering

- To group related or similar documents
- A document is a term vector in the high-dimensional space under the vector space model.

$D_1 = (w_{t1}, w_{t2}, \ldots, w_{tn})$



- Need a measurement of how "close" two documents are

Term Vector Similarity

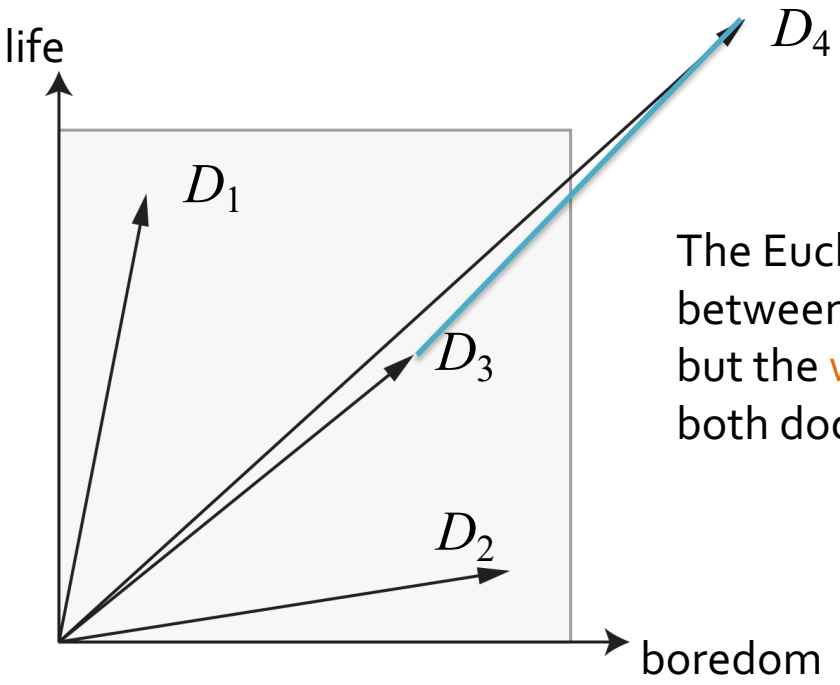
- We shall define the distance between two vectors
- Euclidean distance — consider vectors as points in *n*-dimensional space

$$\text{Let } D_1 = (v_{1,i}) \text{ and } D_2 = (v_{2,i}), \text{ then}$$

$$\text{dist}(D_1, D_2) = \sqrt{\sum_i (v_{1,i} - v_{2,i})^2}$$

- Problem?

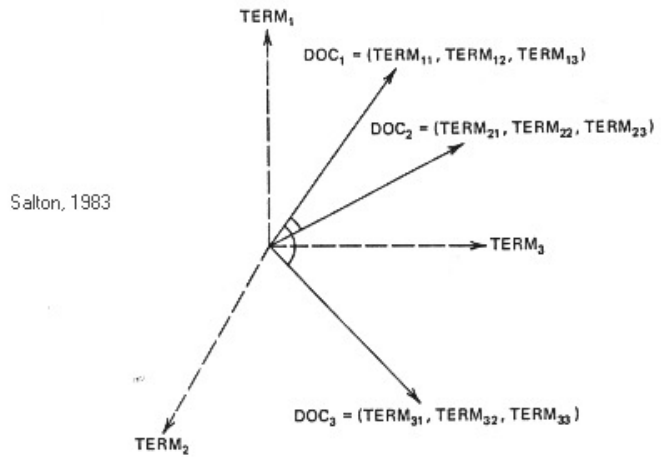
Euclidean Similarity



The Euclidean distance between D_3 and D_4 is large, but the word distribution in both documents are similar.

Cosine Similarity

- **Cosine similarity**: Similarity between documents = cosine of the angle between their vectors



Salton, 1983

Figure 4-2 Vector representation of document space.

Let $D_1 = (v_{1,i})$ and $D_2 = (v_{2,i})$, then

$$\textbf{Similarity}(D_1, D_2)$$

$$= \cos(\theta)$$

$$= \frac{D_1 \cdot D_2}{\|D_1\| \, \|D_2\|}$$

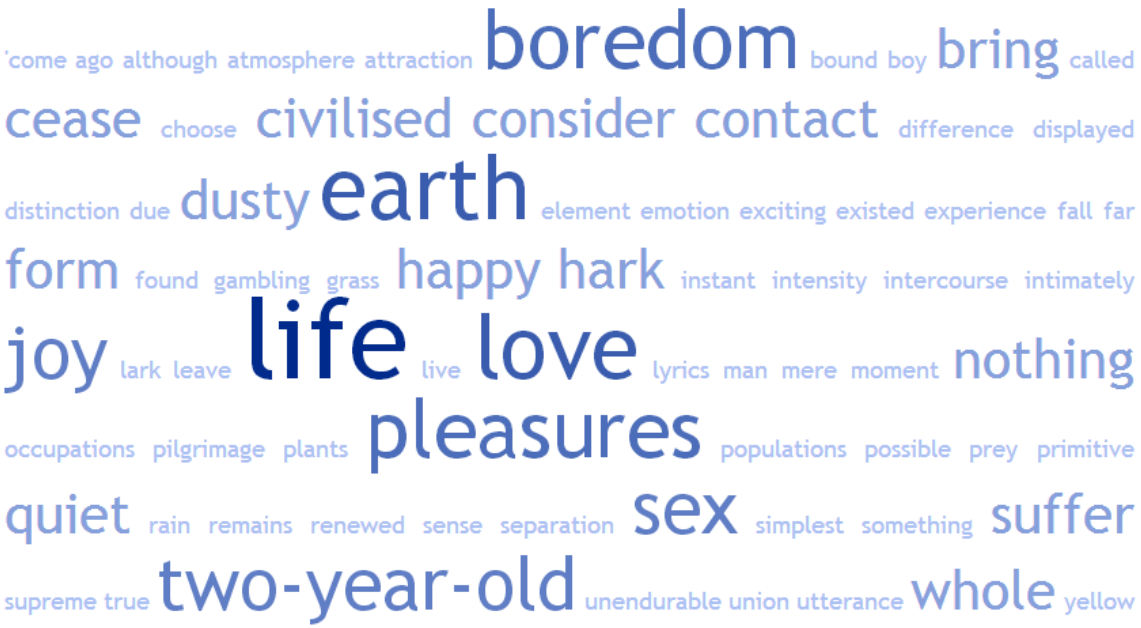
$$= \frac{\sum_i v_{1,i} \cdot v_{2,i}}{\sqrt{(\sum_i v_{1,i}^2)(\sum_i v_{2,i}^2)}}$$

Visualization

- With the vector space model, we may derive patterns and structures from a volume of text which need to be visualized
- Visualization by
  - Words or source text (words / images / phases / paragraphs)
    - tag clouds, bubble cloud, word tree, themail
  - Computed metrics or features
    - arc diagram
  - Models and abstractions, i.e., concepts or themes over documents
    - ThemeRiver, TIARA

Tag Clouds

- A visualization of word frequencies



A paragraph from Chapter 4, "The Conquest of Happiness" by Bertrand Russell.
Tag cloud generated by http://tagcrowd.com/

Tag Clouds

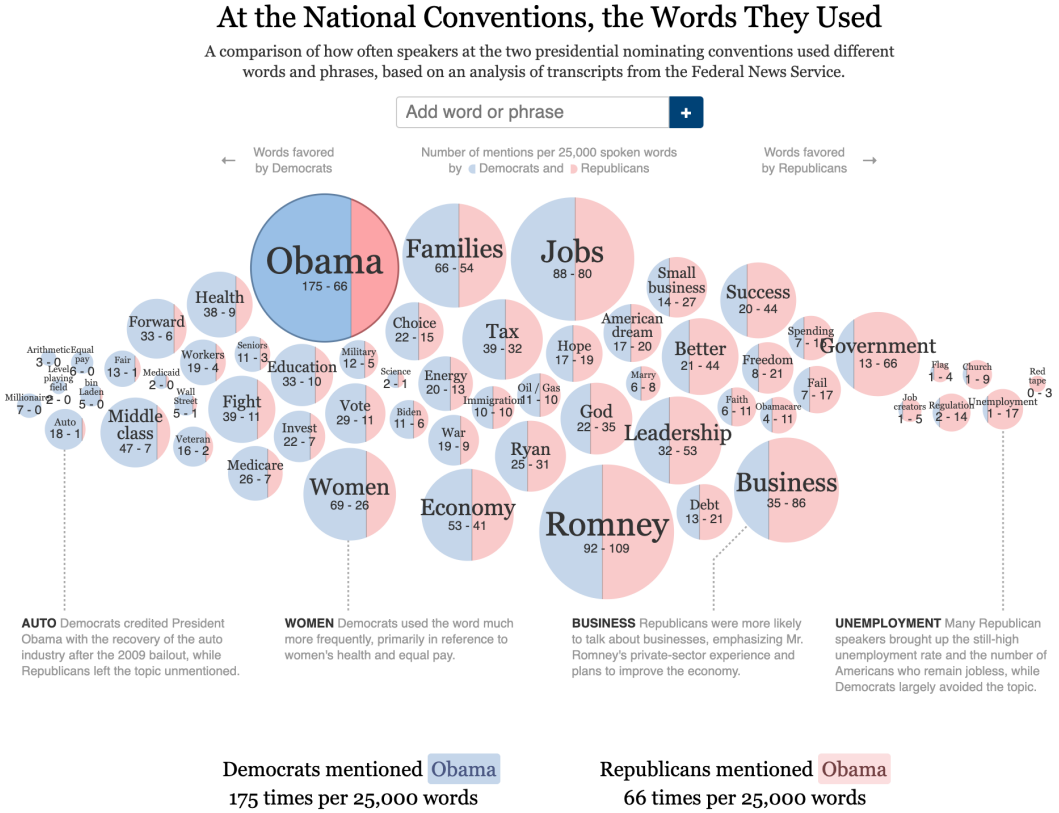
- Size / color determined by frequencies
- Simple and easy to read, good data density
- Potential problems:
  - long words are emphasized over short words
  - Ascenders (e.g., b, d, f, h, k, l, t) and descenders (e.g., g, j, p, q, y) receive unconscious attention
  - Alternatives: bar charts, tables

Tag Clouds



Generated from Wordle [now: http://www.edwordle.net/]

Bubble Cloud

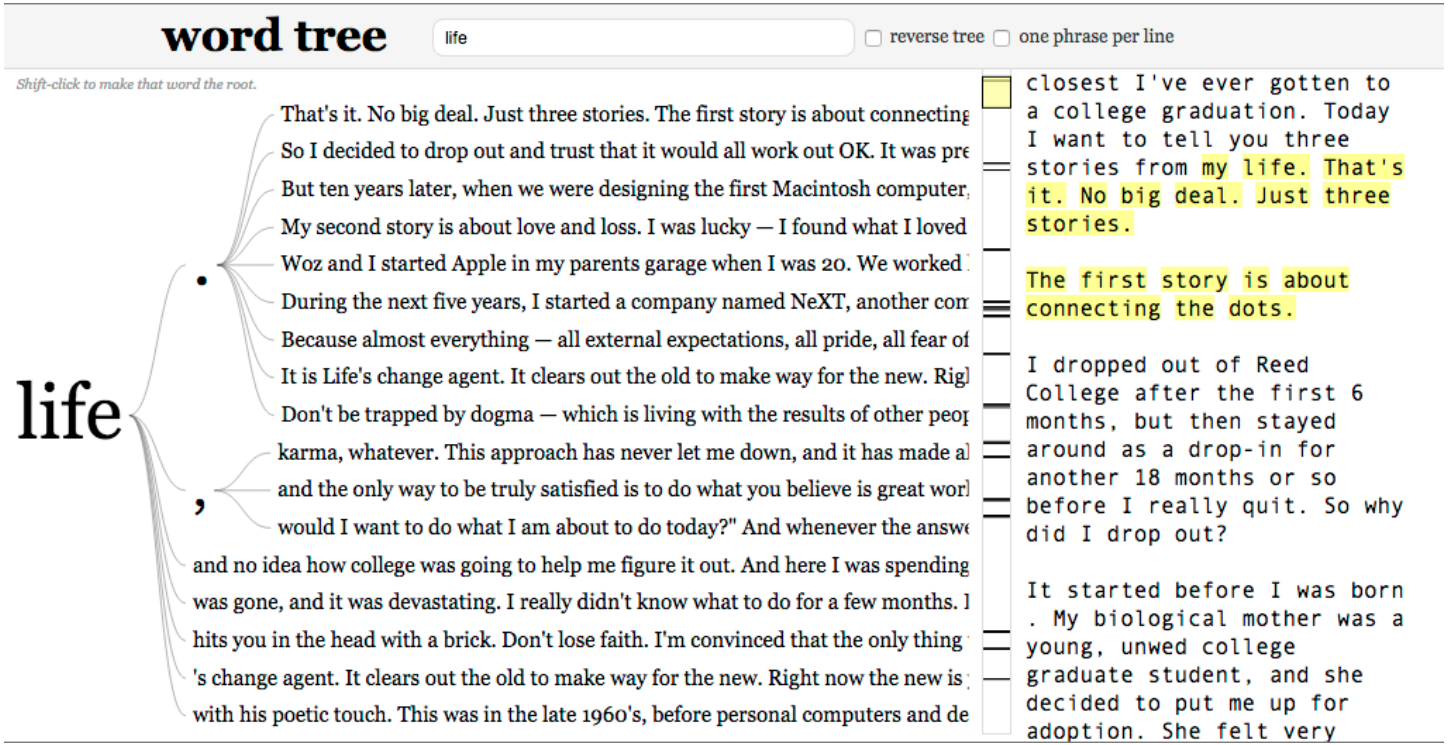


https://archive.nytimes.com/www.nytimes.com/interactive/2012/09/06/us/politics/convention-word-counts.html#Obama

Word Tree

- Visualization of term frequencies, with context
- Document flow visualization
- Root of a tree = user-specified words (***W***)
- Branches = phrases that appear after ***W***
- Similar to the suffix tree

Word Tree



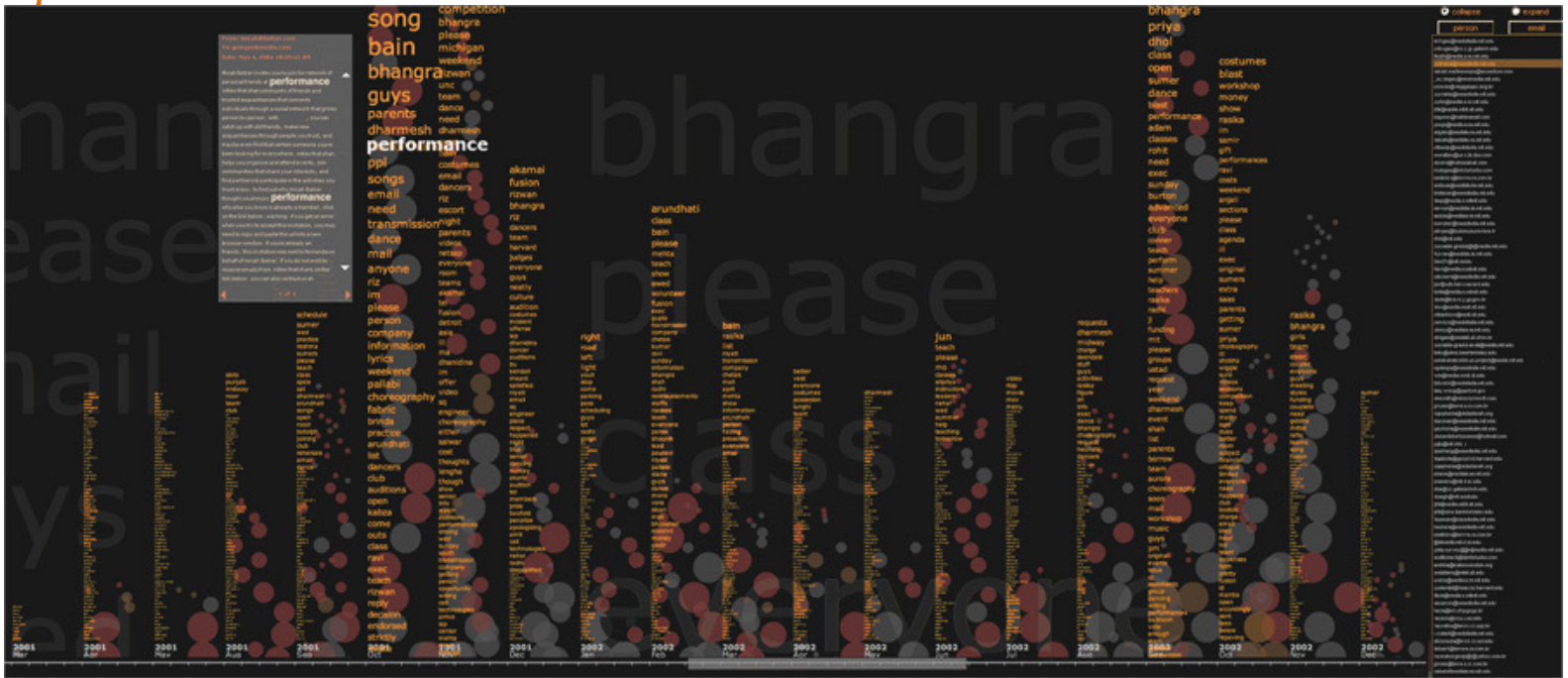
https://www.jasondavies.com/wordtree/

Example: Themail

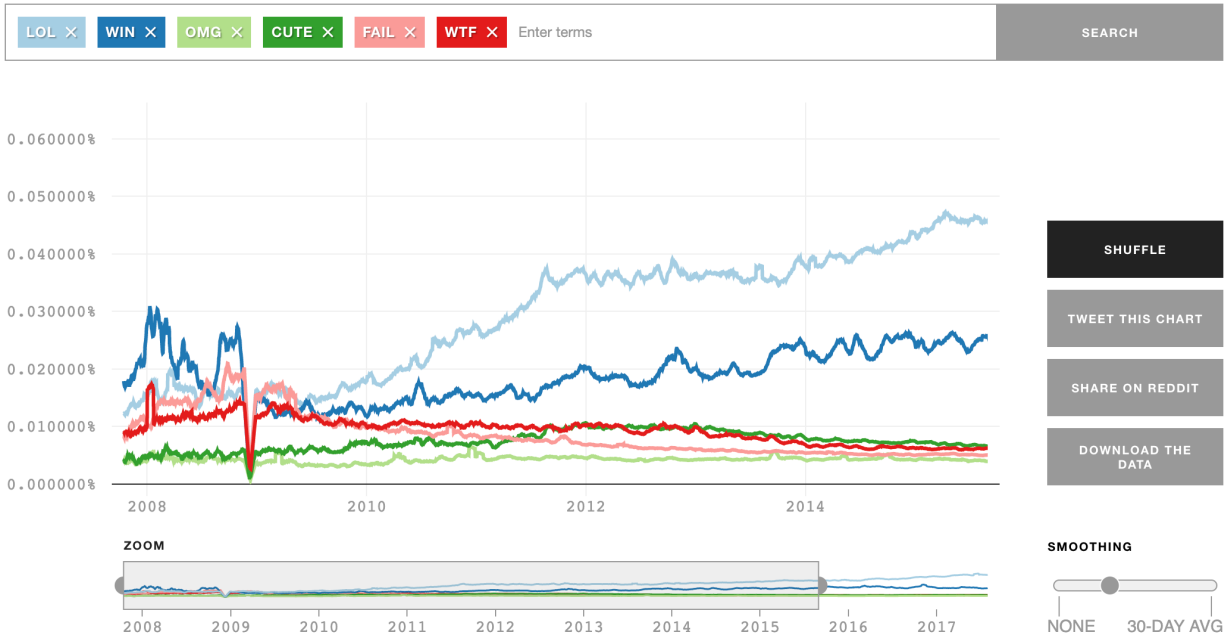
Shows the words that characterize one's correspondence with another and how they change over time

Word importance ↔ Size of words
Email messages ↔ Circles
Length of email ↔ Circle size
Direction of emails ↔ Circle color

[http://alumni.media.mit.edu/~fviegas/projects/themail/]
[https://doi.org/10.1145/1124772.1124919]



Example: How the Internet Talks
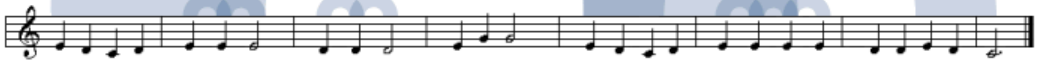


[https://projects.fivethirtyeight.com/...]

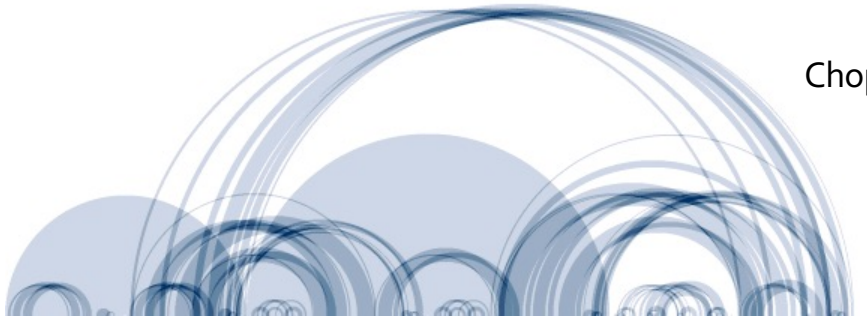
Arc Diagrams

- Show repetitive structures by connecting repeated subsequence with circular arcs
- Good for visualizing musical pieces
- Thickness of arc represents length of the repeated subsequence and height of arc represents distance between the subsequences

The Shape of Songs
[http://www.turbulence.org/Works/song/index.html]



Arc Diagrams



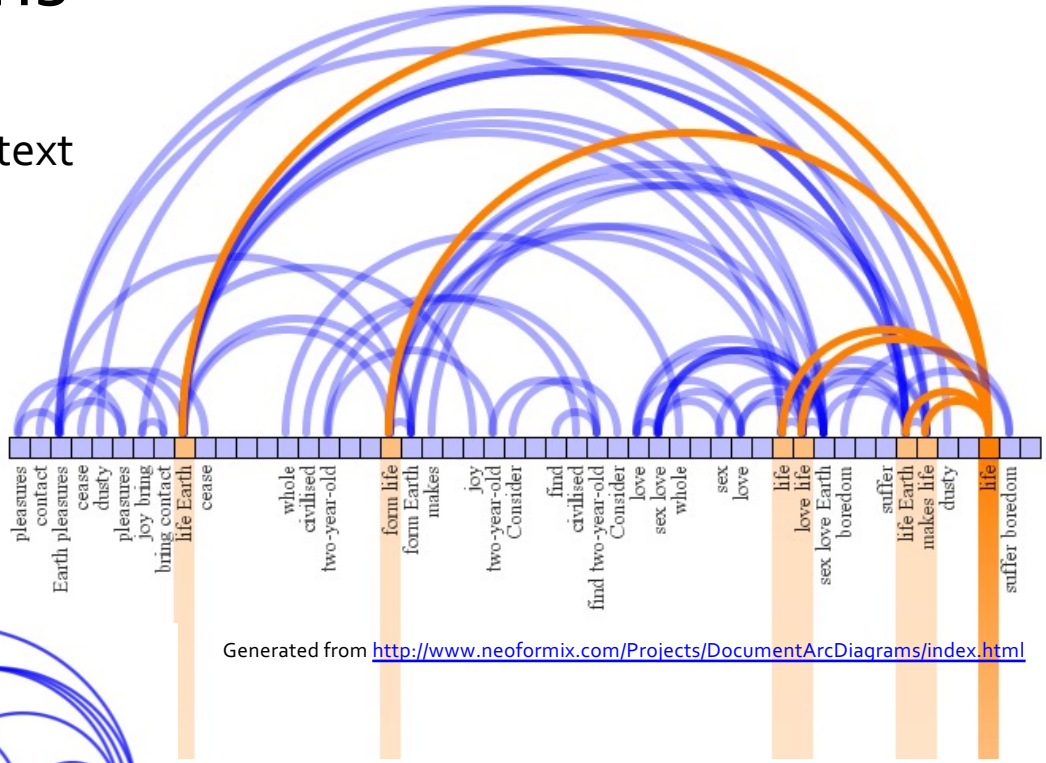
Chopin, Mazurka in F# Minor



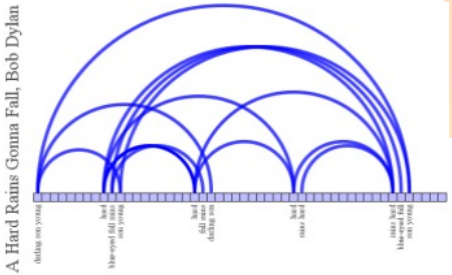
Madonna, Like A Prayer

Arc Diagrams

For free text



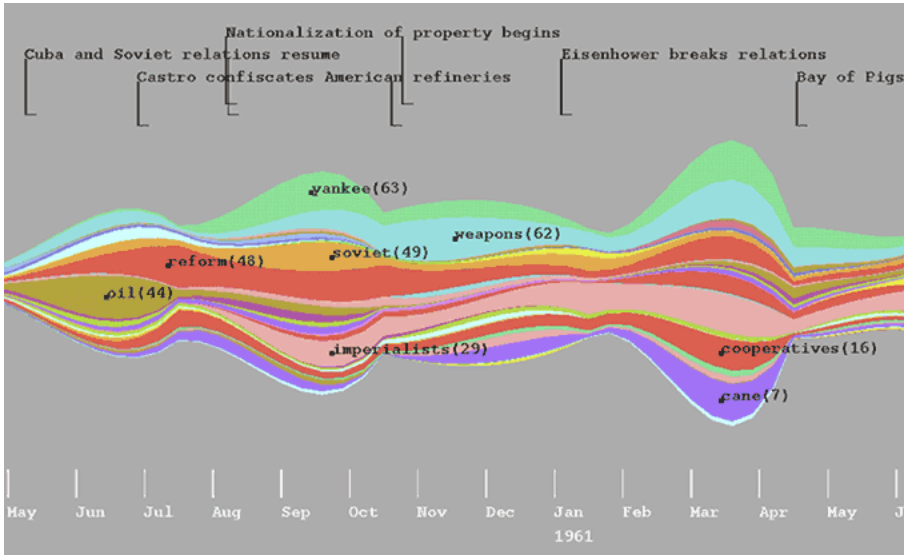
Generated from http://www.neoformix.com/Projects/DocumentArcDiagrams/index.html



Repetitive patterns for lyrics clearly reviewed

ThemeRiver

- Show thematic variations within a corpus over time.
- A river metaphor (direction of flow: time line,
  width: strength of a theme)



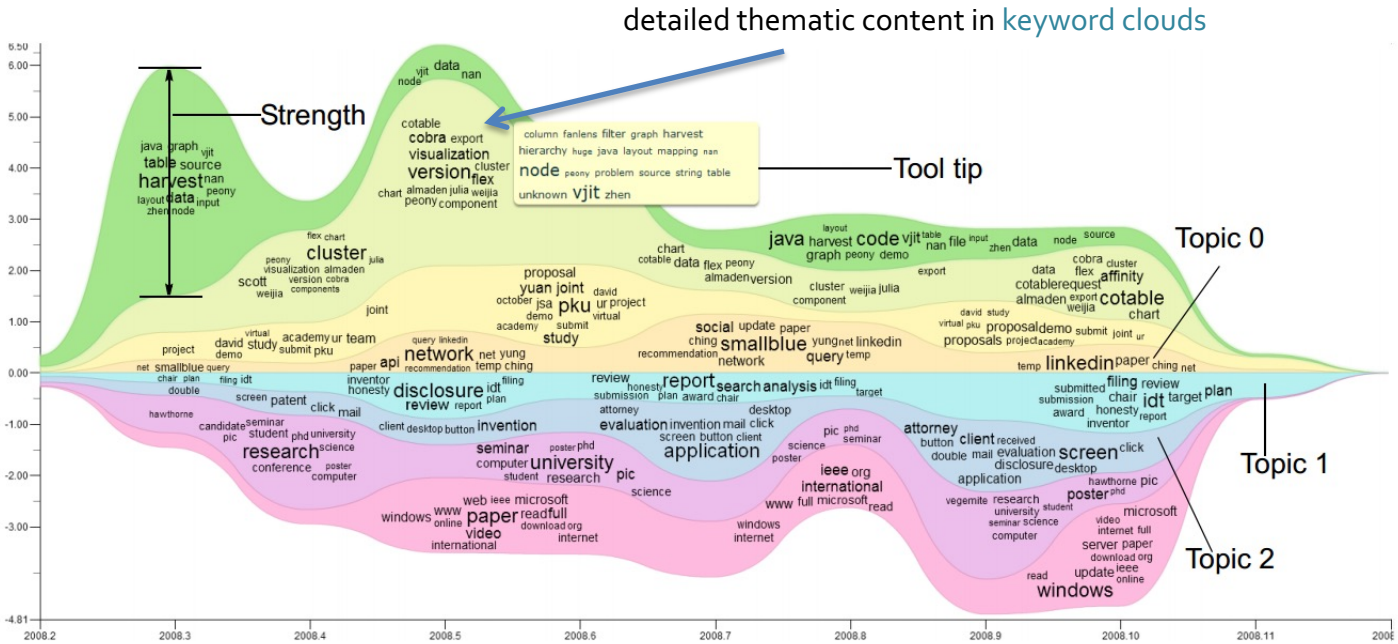
External events

The sudden change of thematic strength following an external event may indicate a causal relationship

https://youtu.be/cWrD5MRWTw8

Tiara

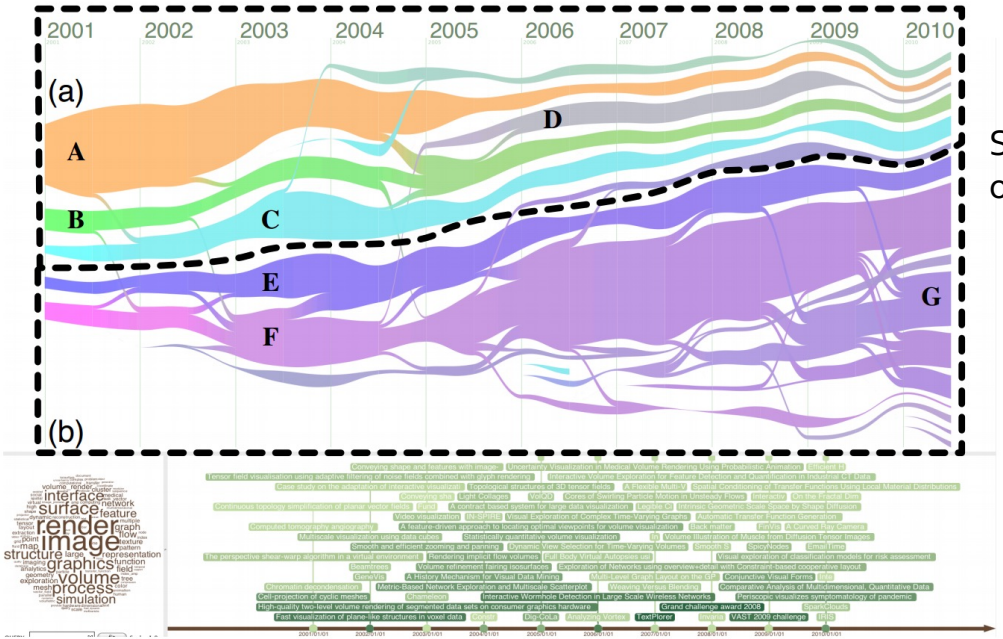
- Illustrates thematic variation over time like ThemeRiver



[Liu et al. "TIARA: Interactive, Topic-based Visual Text Summarization and Analysis", Transactions on Intelligent Systems and Technology, 2012.]

Textflow

- Extract also the connections between evolving topics

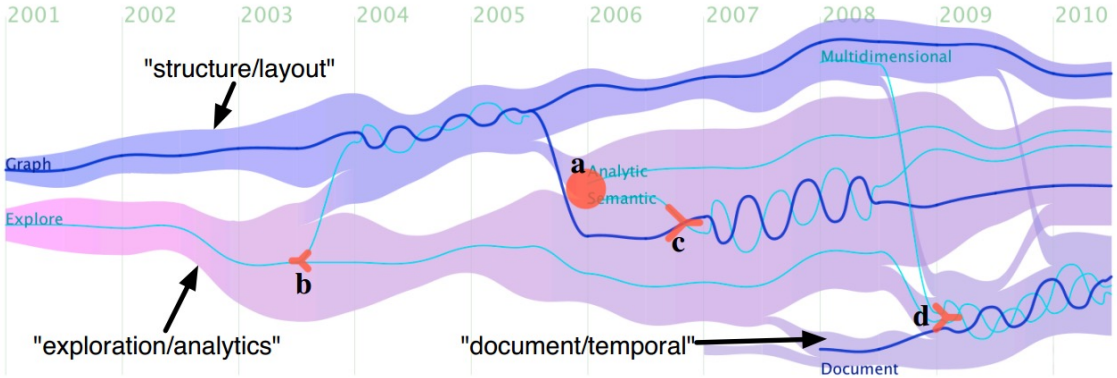


Splitting & merging of topics

[Cui et al. "TextFlow: Towards Better Understanding of Evolving Topics in Text", InfoVis 2011.]

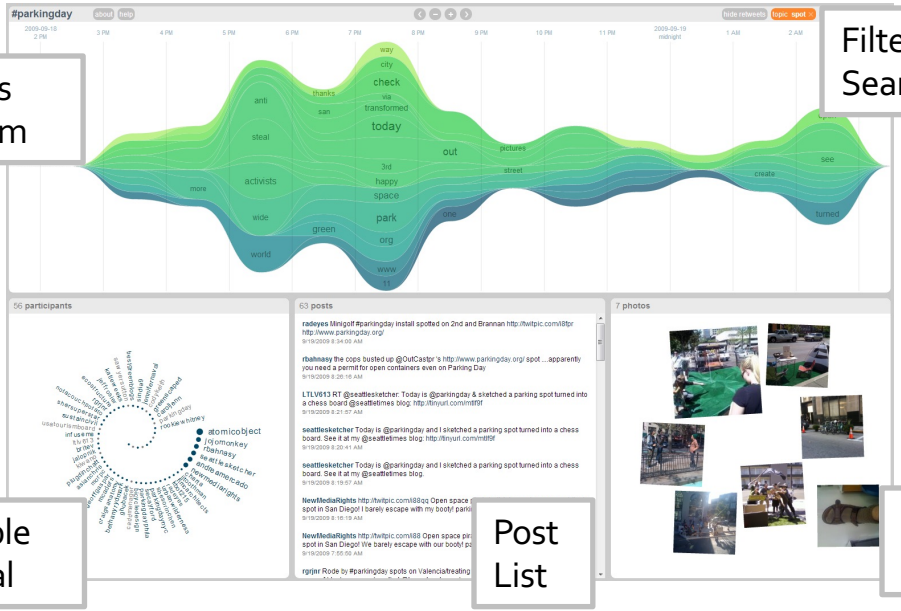
Textflow

- Critical events (topic birth, death, split, merge) in relation to keyword changes



Visual Backchannel

Represent conversation topics in the context of their temporal development together with participants' activity and pictorial impressions



Demo:
http://mariandoerk.de/visualbackchannel/

dot size – number of activities
saturation – originality of posts

[Dörk et al. "A Visual Backchannel for Large-Scale Events ", InfoVis 2010]

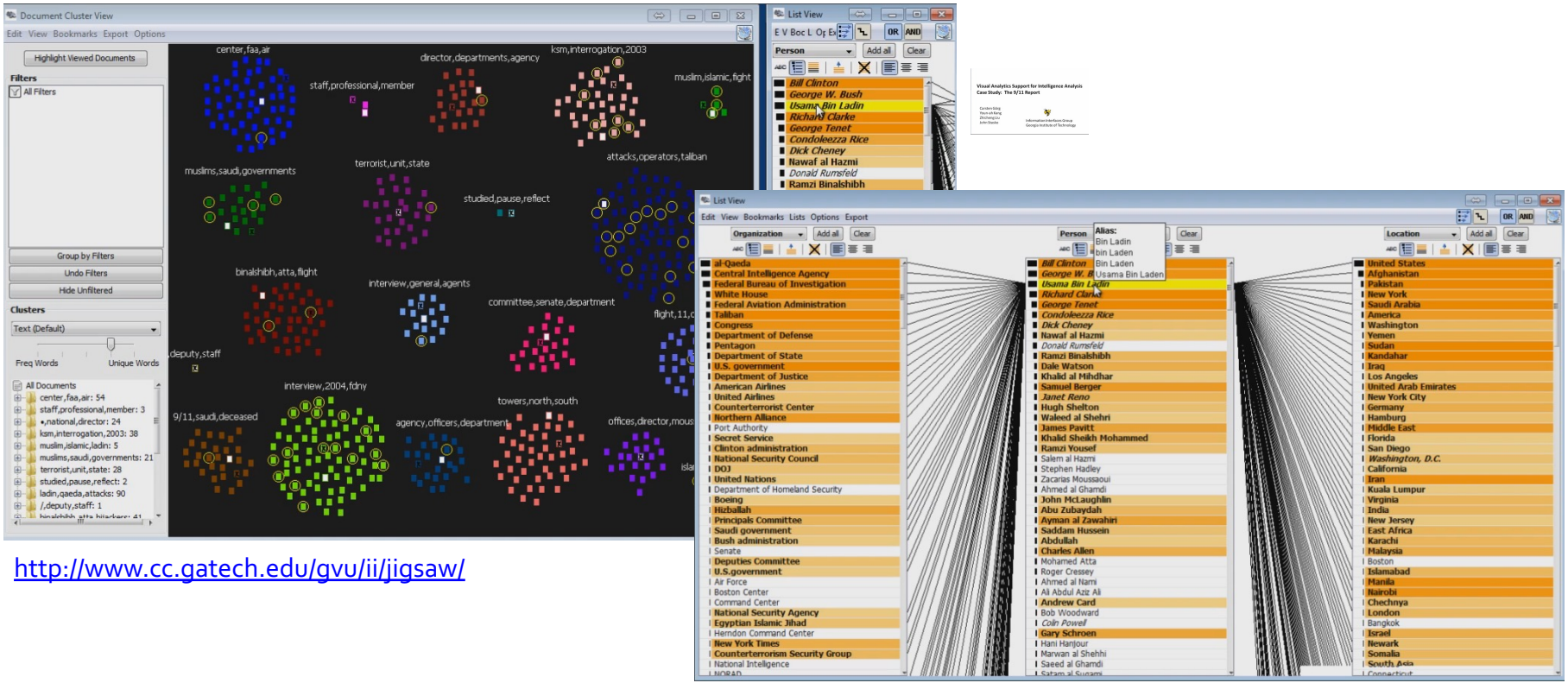
https://youtu.be/OqILOWOblxg

Text Visualization Browser



https://textvis.lnu.se/

Text Analytics with Jigsaw



http://www.cc.gatech.edu/gvu/ii/jigsaw/

Reference

- Christopher D. Manning, Prabhakar Raghavan and Hinrich Schütze, "*Introduction to Information Retrieval*", Cambridge University Press, 2008 [Chapter 6] http://nlp.stanford.edu/IR-book/
- Matthew Ward, Georges Grinstein and Daniel Keim, "*Interactive Data Visualization: Foundations, Techniques, and Applications*", 2010 [Chapter 9]